

Statement of Purpose, Renato D. La Torre, Department of Integrative Biology

I am a Peruvian biologist with experience in genomics, a strong commitment to Neotropical biodiversity research and conservation, and a deep sense of responsibility towards my community. I am interested in characterizing Neotropical biodiversity through space and time and understanding the processes shaping species adaptation. I am particularly drawn to large-scale omics approaches and innovative tools such as environmental and ancient DNA to address interdisciplinary questions. Throughout my academic career, I became proud of the rich Peruvian biological and cultural diversity, and aware of our pressing needs as a country limiting disadvantaged students. Through graduate study, I aim to enrich my tools for making a greater positive impact in my community by generating knowledge and forming highly qualified researchers.

During my undergraduate studies at San Marcos University (UNMSM), I joined diverse research groups to discover and refine my research interests. First, I joined Prof. Mónica Paredes, where I analyzed SNP data from Peruvian family trios to test the association of genetic variants with diabetes risk. I learned statistical concepts and bioinformatic tools, published my findings, and presented in a conference in Brazil. In parallel, I joined Prof. Daniel Oré, with whom I learned programming skills to carry out data mining projects to design lineage-specific primers for endangered Peruvian species, sharing my results in scientific meetings. These experiences honed my laboratory and collaborative skills.

During my senior year, I undertook an internship at the Symbiosis Laboratory led by Dr. Mariana Mateos at Texas A&M University, exploring the interaction between *Spiroplasma* bacteria, *Drosophila melanogaster* and parasitoid wasps. Using qPCR, I revealed wasp-specific patterns of bacterial density within infected flies' larvae throughout their development, suggesting different parasitoid wasp *Spiroplasma*-resistance mechanisms. I shared my findings at the Ecological Integration Symposium. I was captivated by the United States' intellectually-stimulating academic environment.

For my undergraduate thesis, I joined Dr. Gisella Orjeda, then president of the National Institute of Glaciers and Mountain Ecosystems, to study Antarctic lichens using DNA barcoding. Despite pandemic challenges, I analyzed over 300 samples, evaluated the accuracy of DNA-based specimen identification, and inferred phylotypes using multiple delimitation methods. My results contributed to Antarctic lichen representation in the DNA databases, revealed varied identification success among lichen groups, and suggested possible undescribed species. I published my work and presented at the Meeting of the Latin American Lichenologists in Argentina, and received first place in the 1st Latin American Thesis Contest in Mexico.

Convinced of the potential of Neotropical biodiversity, I joined Dr. Orjeda's new genomics laboratory at UNMSM after graduation, helping establish its infrastructure and developing diverse projects. My main focus was the algarrobo, a keystone tree highly-adapted to the Peruvian seasonally-dry forests undergoing unprecedented dieback. In collaboration with Dr. Robin Buell's group at UGA, I led the generation of multiple sources of data and performed the assembly and annotation of the first chromosome-level genome of a Peruvian wild species. Comparative analyses revealed extensive gene loss accompanying genome size reduction, a possible adaptation to its challenging environment. Interestingly, genes functionally-enriched in responses to external stimuli

were expanded, consistent with algarrobo's rapid response to seasonal resource availability. I published my findings and presented at several conferences.

I pursued a master's degree in Ecology and Evolution at NTNU in Norway under Dr. Michael Martin, joining his Holomuseomics group as a student and then hired as research assistant, collaborating in archaeogenomics projects. My thesis built on my previous algarrobo experience to explore the influence of ancient civilizations on genetic diversity through population genomics and archaeogenomics. Analyzing modern, herbarium, and archaeobotanical samples, I characterized the population structure and temporal changes in diversity, revealing a significant reduction in heterozygosity in the almost-extinct southern populations relative to 1,500 years ago. Through my results, I provided a baseline for urgent conservation and highlighted our irreparable influence on Andean ecosystems.

In the Holomuseomics group, I carried out a sedimentary ancient DNA project with Dr. Sarah Martin. My role was key in exploring its potential, as I led the ancient metagenomic analyses in a study involving sediment layers from an archaeological site dating back 3,200 years. Along traditional proxies, we reconstructed local biodiversity and environmental shifts, detecting plant and animal traces, and microbial profiles associated with historical climate events and human activity. Our findings are currently accepted with minor revisions.

I have returned to Peru, where I am responsible for directing our newly-acquired high-performance computing system towards national biodiversity initiatives. One major effort is uniting dozens of stakeholders to digitalize and sequence herbarium specimens across Peru. I also teach the Population Genomics course in the Master's program in Bioinformatics, aiming to strengthen both human capacities and computational resources in Peru.

Understanding past and ongoing adaptation facing unprecedented decline rates is imperative for proactively protecting endangered populations, and requires fundamental research. My work on algarrobo raised questions about how natural and anthropogenic forces shape the adaptability of long-lived species, and prompted my interest in the broader question: "How Neotropical plants respond to current climatic trends?". I am excited about Dr. Moi Expósito-Alonso group's core questions regarding plant adaptation, and their approach tackling those from different angles, particularly through their experimental evolution project GrENE-net. His laboratory values and culture represent a unique and safe growing opportunity. Based on my contact with Dr. Expósito-Alonso, his interest in South American biodiversity and forthcoming access to clean lab facilities, fits greatly with my aspirations, connections and background.

The holistic perspective of the Integrative Biology PhD program, studying life at different levels and from multiple perspectives, aligns perfectly with my academic goals to integrate genomics, ecological context and human history. UC Berkeley's exceptional faculty, research, and commitment to values of community and diversity, represent an inspiring home where I can grow as a scientist, mentor and leader. I am eager to contribute to its vibrant community and to translate my experiences into impactful research that illuminates our understanding of evolution in the Neotropics. Given the opportunity, I am ready to bring dedication, creativity and enthusiasm to every challenge, striving to contribute relevant knowledge at the service of science and society.

Personal History Statement, Renato D. La Torre, Department of Integrative Biology

Through the fog of the Peruvian cloud forest, I saw my father from afar working tirelessly on his coffee cultivation. Although we had never met before and he did not know my name, we looked similar enough for us to know we were related. From 1980 to 2000, violent internal conflict between the “Shining Path” insurgency and the unprepared military forced thousands of Peruvians in the crossfire to emigrate. My family was among those who started from poverty in Lima, hoping to find safety, except my father who stayed risking his life to support us. The strain of separation eventually fractured us, prompting my mother, carrying me in her arms, to start anew, raise her six children by herself, and abandon her professional aspirations.

Growing up in a loving family, I learned from our unity despite hardships and cultivated a deep sense of gratitude. I felt privileged to focus on my education while my siblings endured working after classes. Honoring my family’s sacrifice, I developed a reflective nature, aiming to make the most out of every educational opportunity. My tendency to observe, listen, and try to understand my surroundings were early signs of my inclination towards science. Although we lived in the city’s outskirts, we frequented our acquaintances in the Peruvian Andes and Amazon. These visits taught me to value the hospitality and wisdom of my community and instilled pride in our cultural and biological diversity.

When I met my father for the first time, the image of him alone, old and tired remained burned into my memory. Through my experiences, I have met many Peruvians that, similar to my siblings and parents, had their dreams cut short. Despite its rich diversity, Peru’s social instability and lack of opportunities in rural areas exacerbate systemic inequality. Aware of our reality, my gratitude transformed into a deep sense of responsibility toward my community. Knowledge is a privilege that comes with the duty to act, and I am determined to have an impact in Peru bringing opportunities to reduce the unfair disadvantage faced by many talented people.

As the top Peruvian public university, San Marcos University congregates diverse students from all over the country. As I progressed my undergraduate studies, I felt at home and quickly loved the academic environment: learning, discussing and research. During my early years, I adapted to economic hardship with initiative. For example, I translated scientific papers to my peers charging 0.2 USD for each page to support my education while improving my English skills. I organized a network of contacts to share used laboratory consumables for training, forming long-lasting friendships. These experiences seeded my notion of the value of building inclusive and collaborative scientific communities. My international experiences broadened my perspective on diversity, as I felt welcomed to bring my uniqueness to different vibrant and inclusive academic environments, particularly in the US.

My academic career is marked by perseverance and conscientiousness. After finishing my undergraduate studies during the pandemic, while simultaneously protecting vulnerable family members, I took the time to reflect on my next steps. Having a broad interest in research, and certain of my eagerness to contribute to a relevant issue in Peru, I chose to work in my country while refining my career interests before pursuing a graduate degree. Although public insecurity and low remuneration were challenging, working on the algarrobo project became one of the most meaningful experiences of my life. I saved

money working as a kitchen assistant and learned new bioinformatic skills before being able to lead the genomics project. I gathered the perspectives of stakeholders and families affected by biodiversity decline, whose hopes reaffirmed my purpose of contributing to my community generating knowledge.

Taking the risk to explore new academic and personal horizons, I took the opportunity to experience a European academic environment before a PhD. Studying abroad was challenging, especially adapting to the Arctic climate, but I grew to appreciate the Norwegian values of tolerance, egalitarianism, and outdoor life. Far from my family, I learned to understand diverse backgrounds and effectively communicate my own experience to connect with my peers. My involvement in many exciting projects resulted in my grades lowering the second semester. Instead of discouraging myself, I reassessed my priorities and focused on my thesis, ultimately earning the highest grade. The beautiful Scandinavian landscapes ingrained my love for nature, and drove me to reflect on my history, objectives and goals. Realizing that I could not steer away from my mission in Peru was decisive in my hesitation about continuing an arctic-focused PhD.

I seek to focus my research in the Neotropical largely-understudied biodiversity, convinced of its enormous potential and its relevance to rural families. Back in Peru, I contribute to forming capable researchers by guiding students in their thesis projects, sharing my expertise in the laboratory and classrooms, and managing my university's new HPC system, which will expand computational access for research groups nationwide. I also volunteer in environmental and social campaigns, where I challenge current stereotypes by projecting a friendly image as a scientist. These roles strengthen my conviction that research, mentorship and community-building must be interconnected.

At my core, I am shaped by gratitude, responsibility and pride into a resilient, service-oriented and reflective person, strongly rooted with my community. My experiences align with my aspiration to become a principal investigator who not only advances scientific knowledge but also expands research capacity and educational access in Peru. In particular, I aim to instill in students my love for our nature and hope for our society, and to expand their range of possibilities, just as mentors and professors expanded it for me. I am convinced that a PhD is the next step in my career, and I believe that UC Berkeley's values of public service, innovation, and inclusive excellence strongly resonate with my own. UC Berkeley offers a unique environment where I can grow into the scientist and leader my community deserves. Given the opportunity, I am committed to contributing meaningfully to a more equitable and sustainable world.